

Journal Entries

Date: 27/07/2026 – 02/08/2026

Progress:

Phase 3 took 1 day to finish. Phase 4 and 5 are completed as well. Small obstacles with how much time it took, but I think I am quite ahead of my plan.

Result:

Phase 3 showed that CatBoost and XGBoost outperformed DecisionTree once imbalance handling was applied => the ranking from Phase 2 flipped. I took the top three combinations forward to Phase 4 for hyperparameter tuning with Optuna (50 trials each, 5-fold stratified CV). The best-performing model was XGBoost + RandomOversampler with CV macro-F1 = 0.5188. On the sealed test set in Phase 5, this held up at macro-F1 = 0.5194, nearly double the naive floor of 0.28. Fatal-class recall improved from 22% (DT baseline) to 50% (final tuned) => the biggest single achievement of the project. TreeSHAP identified Accident_Type as the dominant feature (~3× the next), with per-class analysis revealing that Day_Phase disproportionately drives fatal predictions.

Next Step:

- Start with the last phase, which is the dissertation.
- Check all the requirements
- Email my supervisor
- Start writing notes and answering possible questions.